Technical University of Crete

School of Electrical and Computer Engineering

Course: Reinforcement Learning and Dynamic Optimization

Programming Assignment 2 (Second part)

Algorithms to implement (Learner)

UCB. At each time step t, the learner selects the price arm i_t from the discrete set of K price levels $\mathcal{P}$ according to the following rule:

$$i_t = \arg\max_{i \in \{1,\dots,K\}} \text{UCB}_i(t) \text{ , with } \text{UCB}_i(t) = \hat{\mu}_i(t) + (p_{max} - p_{min}) \sqrt{\frac{2 \log T}{N_i(t)}}$$

where $\hat{\mu}_i(t)$ is the empirical mean reward of price $p^{(i)}$ up to time t, $N_i(t)$ is the number of times price $p^{(i)}$ has been selected, and T is the total time horizon. The confidence bonus is scaled by $(p_{max} - p_{min})$, representing the possible reward in the given price range $[p_{min}, p_{max}]$.

MW in bandit mode. At each round t, the learner selects price i from a distribution incorporating an exploration parameter ϵ:

$$q_i(t) = (1 - \epsilon) \frac{w_i(t)}{\sum_{j=1}^{K} w_j(t)} + \frac{\epsilon}{K}$$

After observing reward r_t, it computes an unbiased, normalized loss estimate: $\hat{\ell}_i(t) = \dfrac{1 - (r_t / p_{max})}{q_i(t)}$

Only the weight for the chosen arm is updated using the learning rate η: $w_i(t+1) = w_i(t)(1-\eta)^{\hat{\ell}_i(t)}$

Competitor policy: myopic best response

Against UCB. Because UCB is a deterministic algorithm, we assume the competitor can accurately predict the exact price p_t posted by the learner in round t. The competitor selects its price q_t to maximize its expected reward:

$$q_t = \arg\max_{q \geq 0} [\mathbb{E}[R_c(q)]] = \arg\max_{q \geq 0} [q \times \Pr(\text{choose competitor})] = \arg\max_{q \geq 0} \left[q \times \left(1 - \frac{1}{1 + \exp(\beta(p_t - q))} \right) \right]$$

Against MW. Because the MW algorithm is stochastic, the competitor cannot know p_t in advance. However, assuming it knows the learner's probability distribution $\vec{q}(t)$ over the discrete set of prices $\mathcal{P}$, the competitor maximizes its expected revenue by computing the expectation across the learner's distribution:

$$q_t = \arg\max_{q \geq 0} [\mathbb{E}[R_c(q)]] = \arg\max_{q \geq 0} \sum_{k=1}^{K} q_k(t) \left[q \times \left(1 - \frac{1}{1 + \exp(\beta(p^{(k)} - q))} \right) \right]$$

where $q_k(t)$ is the probability that the learner pulls arm k in round t.

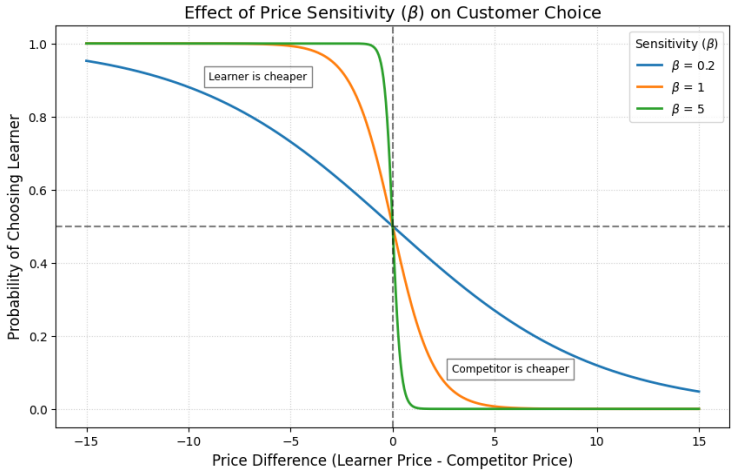
Price war game across different price sensitivities

To evaluate the robustness of the UCB and MW algorithms against a myopic best response competitor, we simulated the price war game across three qualitatively different customer price sensitivity regimes: low ($\beta = 0.2$), medium ($\beta = 1$), and high ($\beta = 5$). All experiments were averaged over 20 Monte Carlo epochs with a time horizon of $T = 1000$, using the same discrete set of $K = 5$ prices.

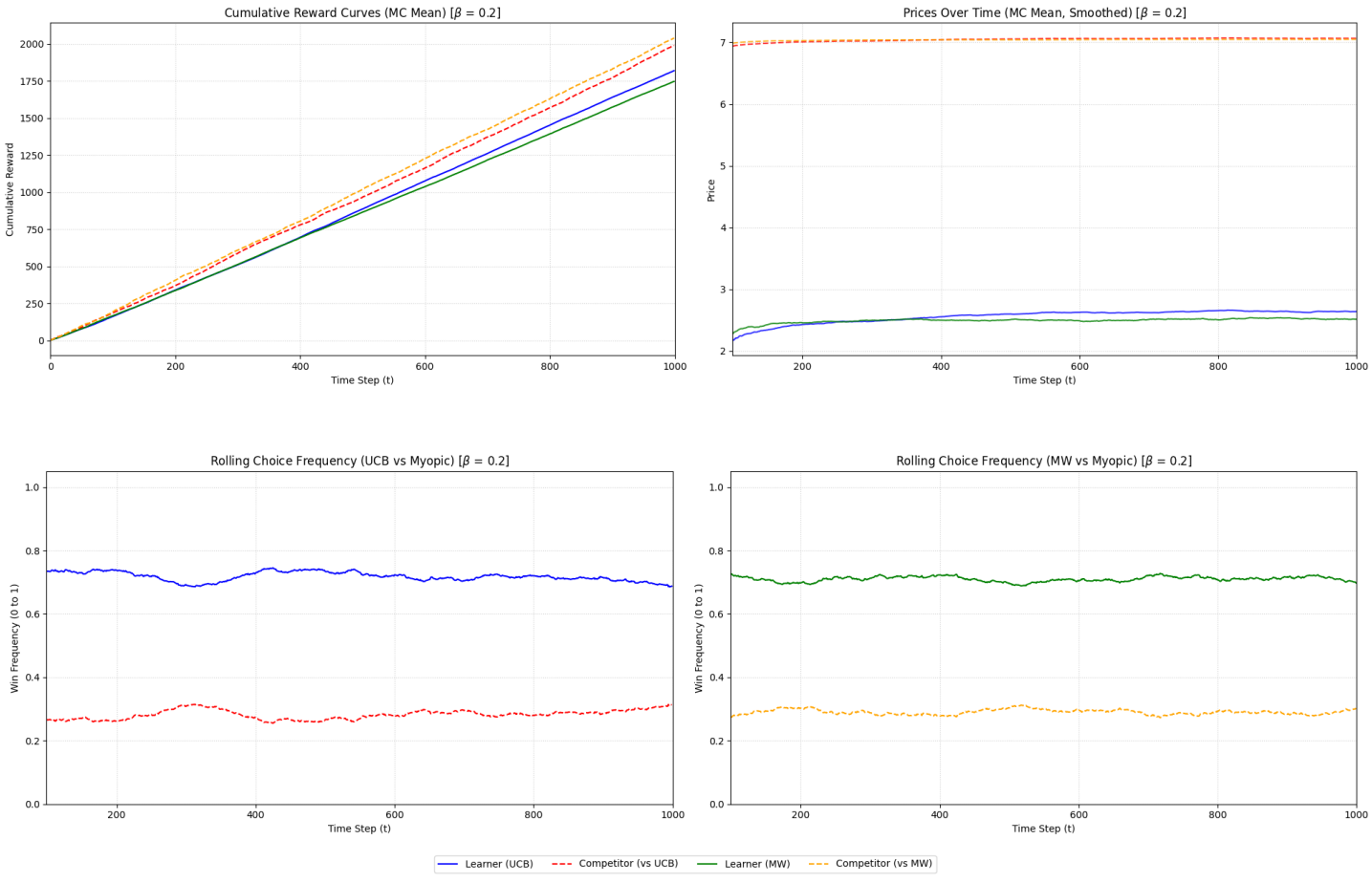
The customer chooses the learner with probability:

$$\Pr(\text{choose learner}) = \frac{1}{1 + \exp(\beta(p_t - q_t))}$$

A larger β means customers respond more strongly to price differences and therefore choose the cheaper option more decisively, as also illustrated by the figure on the right.



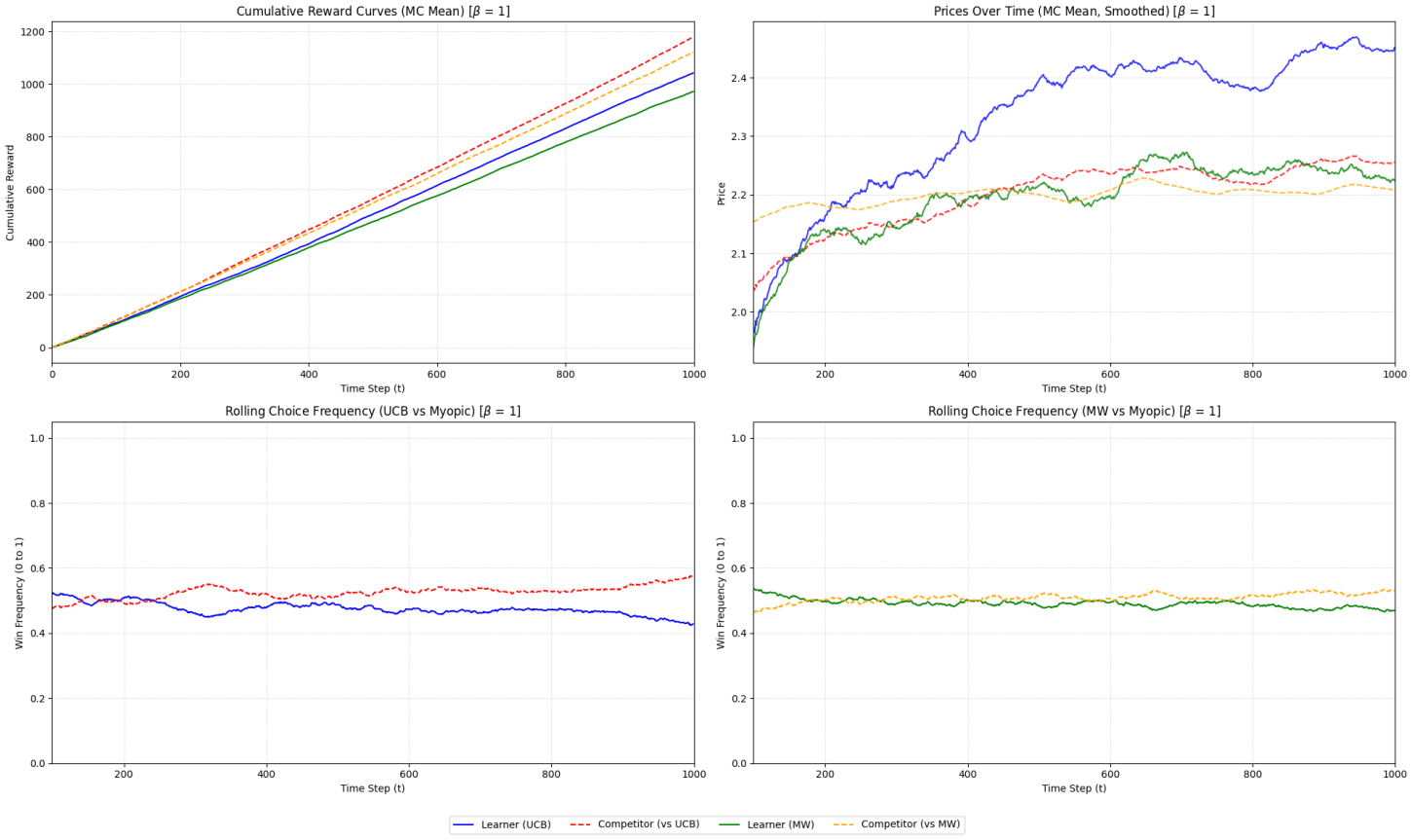
Regime 1: Low Price Sensitivity ($\beta = 0.2$)



Algorithm	Learner Final Reward	Competitor Final Reward
UCB	1818.38 ± 49.73	1989.09 ± 106.12
MW	1745.56 ± 53.48	2039.34 ± 89.96

In a low-sensitivity market, the competitor sets high prices to maximize margins, effectively conceding market share. Both UCB and MW capture roughly 70% of the volume by offering lower prices. Since strict undercutting isn't required, UCB's predictability goes unpenalized, allowing it to match MW.

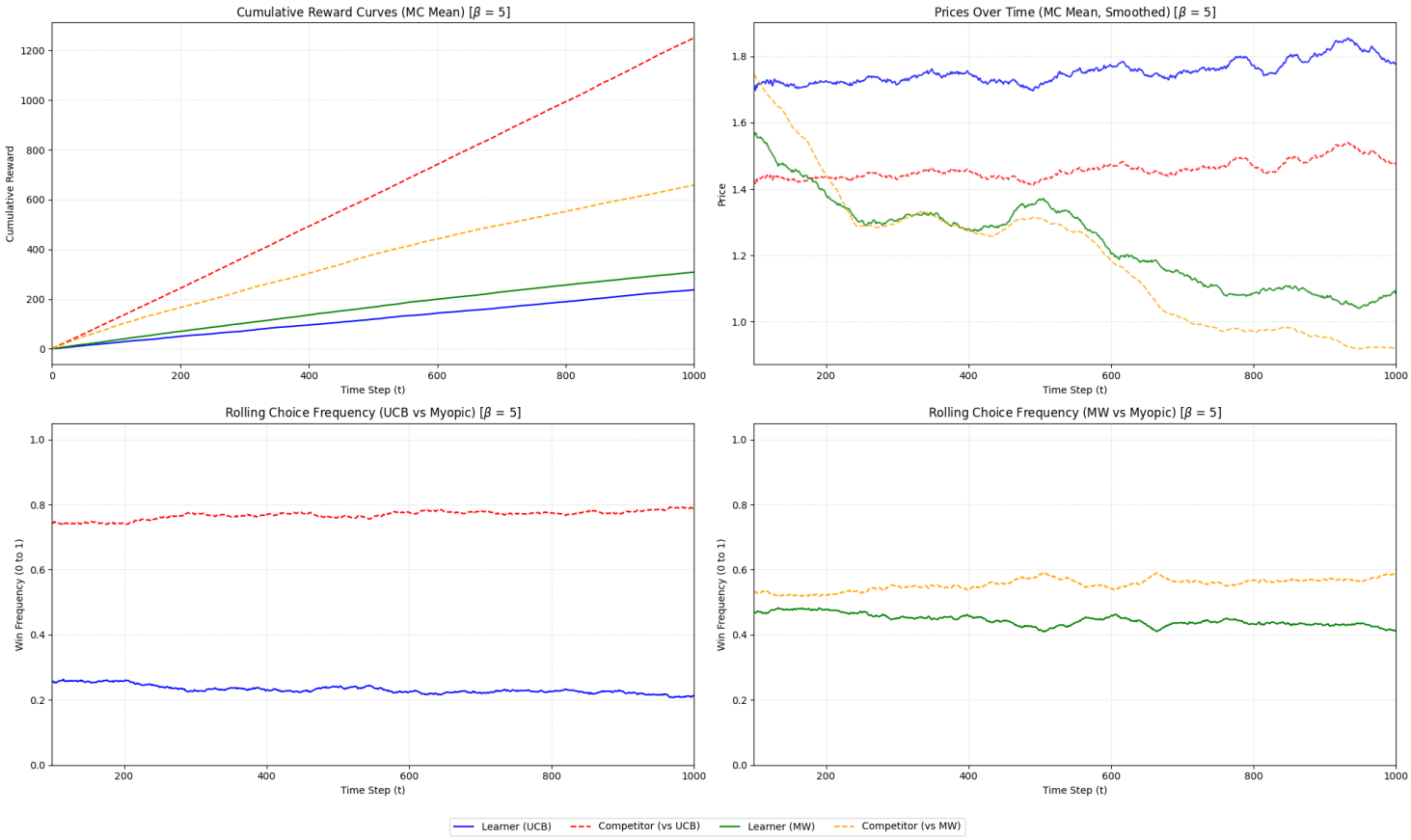
Regime 2: Medium Price Sensitivity ($\beta = 1.0$)



Algorithm	Learner Final Reward	Competitor Final Reward
UCB	1041.53 ± 41.29	1178.05 ± 29.00
MW	970.94 ± 38.80	1120.15 ± 83.80

As the market becomes more rational, the dynamics shift into a highly contested price war. The competitor can no longer afford to set maximum prices and is forced to actively track and undercut the learners. Prices exhibit a gradual upward trend rather than immediate convergence, starting near 2.0 and climbing toward 2.2 - 2.4 as the horizon progresses. Throughout this adjustment, the win frequencies for all agents hover around 50%, though the myopic competitor establishes a substantial cumulative reward advantage.

Regime 3: High Price Sensitivity ($\beta = 5.0$)

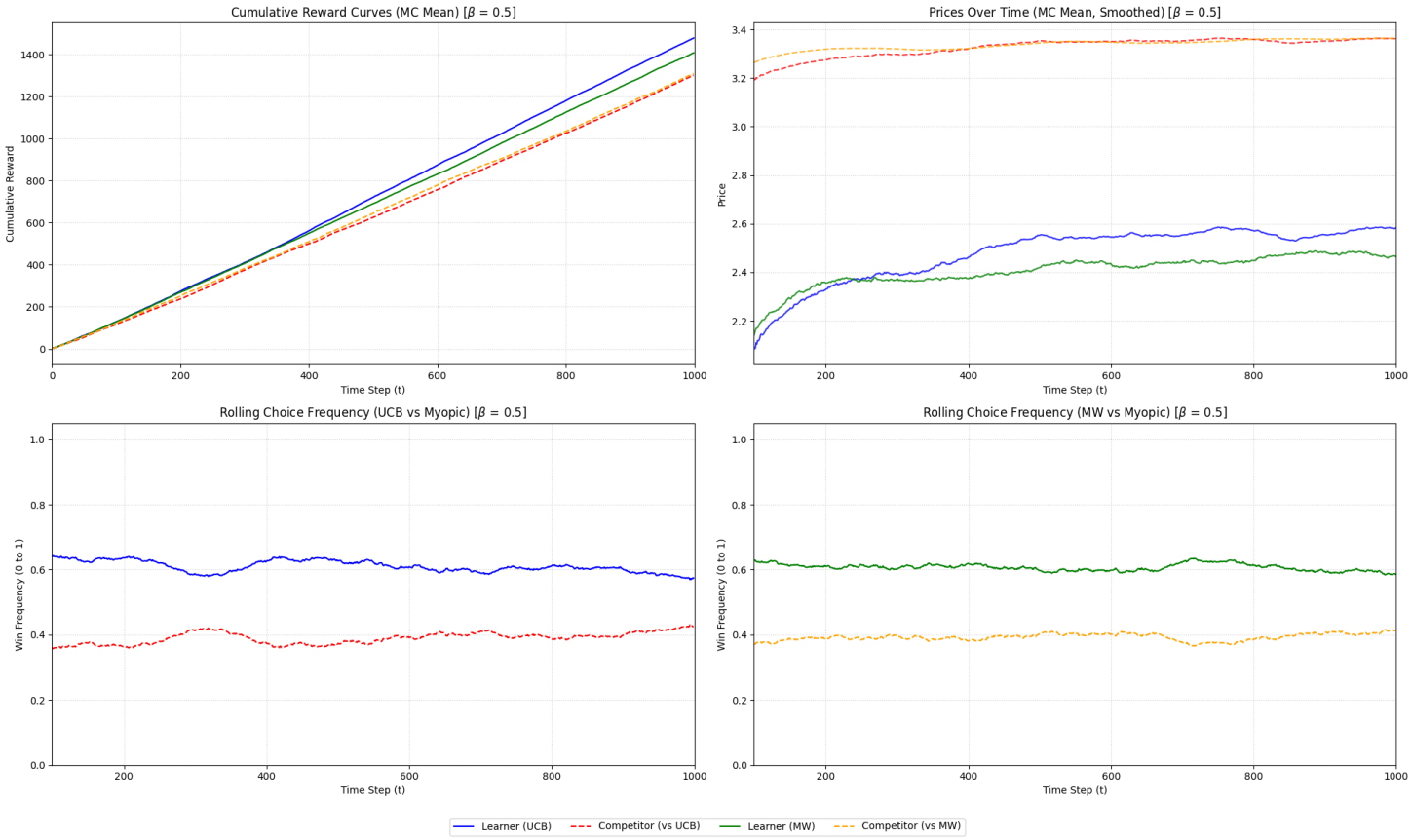


Algorithm	Learner Final Reward	Competitor Final Reward
UCB	236.94 ± 18.41	1249.62 ± 27.17
MW	308.10 ± 52.48	658.37 ± 189.86

In a high-sensitivity market, UCB's deterministic strategy allows the competitor to easily predict and consistently undercut its static price ($\sim$1.7), dropping the learner's win rate to roughly 20%. Conversely, MW proves much more robust, initially keeping pace with the competitor before a noticeable gap in cumulative rewards develops by the end of the horizon. Its stochastic strategy prevents the competitor from perfectly anticipating prices, maintaining a 45% win frequency and limiting the competitor's gains relative to UCB.

Regime 4: Learner Win Scenario ($\beta = 0.5$)

We should point out that for $\beta = 0.5$ the competitor is no longer conceding market share. Compared to the $\beta = 0.2$ case, it lowers its prices from 7.0 to 3.3 and thereby recovers about 10% of the total volume; however, this comes at the cost of lower cumulative reward, leaving the learner ahead overall.



Algorithm	Learner Final Reward	Competitor Final Reward
UCB	1479.18 ± 39.30	1301.98 ± 40.01
MW	1408.53 ± 47.51	1309.55 ± 50.83

Overall, as we move from lower to higher values of β, we observe a clear change in the competitor's myopic pricing behavior. For low β, where customers are relatively insensitive to price differences, the competitor tends to post higher prices, prioritizing margin over volume in the hope of achieving a higher overall reward despite losing some market share. In contrast, for higher β, when customers respond more strongly to price gaps, the competitor consistently adopts a more aggressive strategy and repeatedly undercuts the learner in order to capture demand. This reflects the fact that the competitor's myopic best response becomes increasingly volume-driven as price sensitivity rises.